MARK SCHEME · TEACHER COPY

10BR · Chapter 6: Automated and Emerging Technologies

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

For teacher / admin use only. Do not give to students before they have attempted the paper.

Mark scheme extracts © Cambridge University Press & Assessment.

Question 1 · May/June 2023 Paper 11, original Q9 (a),(b),(c) [5 marks]

Question	Answer	Marks
1(a)	Any **two** from: • It has a **mechanical** structure/framework • It has **electrical** components // by example	2
1(b)	Any **two** from: e.g. • Employees don't need to lift heavy furniture • Employees can be protected from dangerous tasks • Employees can utilise their skills in other tasks • Employees don't need to perform repetitive/mundane tasks	2
1(c)	Any **one** from: e.g. • Expensive to **install/purchase/setup** • High ongoing costs/maintenance costs • May deskill the workforce • If they malfunction, production may stop	1

Question 2 · May/June 2023 Paper 12, original Q9 (a),(b) [5 marks]

Question	Answer	Marks
2(a)	**Three** from: • Rule base • Knowledge base • Interface	3
2(b)	Any **two** from: • It makes decisions • … by **applying** the rules/logic to the facts/knowledge … • … to provide a result/diagnosis	2

Question 3 · Oct/Nov 2023 Paper 12, original Q10 (a),(b) [7 marks]

Question	Answer	Marks
3(a)	Any **one** from: – The ability to learn/adapt // machine learning abilities – The collection of data and the rules for using that data – The ability to reason // has problem solving abilities // makes predictions – Simulates intelligent/human behaviour – Analyses patterns	**1**
3(b)	Any **six** from: – It has an interface … – … used to input data/view output – It has a knowledge base – It has a rule base – It has an inference engine – Applies the rule base to/and the knowledge base to provide output/diagnosis/result/solution/decision – Decides what to ask next based on the data input	**6**

Question 4 · Oct/Nov 2023 Paper 13, original Q10 (a),(b) [9 marks]

Question	Answer	Marks

4(a)	**One** mark for each correct term, in the correct order: – Knowledge base – Inference engine – Rule base // knowledge base – Knowledge base // rule base – Interface	**5**
4(b)	Any **four** from: e.g. – It is a form of artificial intelligence – Means it can adapt/change (its own processes) // It can edit its own algorithms – It can edit its own data – It can be trained – … this can be supervised/unsupervised – … meaning it can learn with/without human interaction – Analyses patterns and stores **successful**/**unsuccessful** results … – … to influence future decisions – (Supervised) means a user tells the system the input and output – (Unsupervised) means the system is given the input and needs to work out the output	**4**

Question 5 · May/June 2024 Paper 12, original Q10 (a),(b) [7 marks]

Question	Answer	Marks
5(a)	Any **two** from: • The **simulation** of intelligent behaviours by computers • A collection of data and the rules for using that data • Has the ability to reason • Has the ability to learn/adapt	**2**

5(b)	Any **five** from: • The user will enter data into the **interface** // The user will plug the car into an **interface** // The user is given questions using the **interface** • The **inference engine** will decide which questions to ask • … by using the previous answers given • The **inference engine** will decide on a diagnosis • … by **comparing the data/answers** entered to the **knowledge base** and **rule base** • … by calculating which option is most likely if there are multiple • The **interface** will output the **diagnosis/result** • The **explanation system** shows **how** the diagnosis was reached	**5**

Question 6 · Oct/Nov 2024 Paper 11, original Q9 (b) [1 marks]

Question	Answer	Marks
6(a)	Machine learning	**1**

Question 7 · Oct/Nov 2024 Paper 12, original Q8 (b),(c) [6 marks]

Question	Answer	Marks
7(a)	• Knowledge base • Rule base • Inference engine	**3**
7(b)	Any **three** from: For example: • It could **gather data** during **vacuuming** • … and adapt its **own processes** • … such as where obstacles are placed in the room • … such as where dirtier areas are • … such as a path through a room • … such as the shape of a room • … such as the most efficient route to vacuum a room • … so, it knows areas to avoid/concentrate on/use different cleaning tools	**3**

Question 8 · Feb/March 2025 Paper 12, original Q6 (a),(b) [6 marks]

Question	Answer	Marks
8(a)	Any three from: • Collection of data • Rules for data • Ability to reason • Ability to adapt	3
8(b)	Any three from: Examples • **Changes** its data/rules • … according to feedback/results • If a result does not give a suitable response this is recorded • … next time the result is changed • … to provide more accurate/relevant results in future • Supervised/unsupervised learning	3

Question 9 · May/June 2025 Paper 11, original Q8 (a),(b),(d),(e) [5 marks]

Question	Answer	Marks
9(a)	Any **one** from: • It has electrical components. // by example • It is programmable. • It can move.	1
9(b)	Any **one** from: • Permanent **instructions/software** that are programmed into/stored in the ROM. • **Instructions/software** that allow hardware to be controlled/managed. • **Instructions/software** that provides the operating system with a platform to run on.	1

9(c)	Any **two** from: Examples: • Workers don't need to lift **heavy** machinery • The workers are less likely to injure themselves • The workers don't need to be in any dangerous environments in the factory • The workers don't need to perform boring/repetitive tasks such as moving car parts • Workers can focus on having higher expertise skills • Create new jobs for workers to maintain the robots.	**2**
9(d)	Machine learning	**1**

Question 10 · Oct/Nov 2025 Paper 13, original Q6 (a),(b) [6 marks]

Question	Answer	Marks
10(a)	Any **three** from: • Simulation of intelligent behaviours by computers • Collection of data and rules for using that data • A computer that has the ability to learn • A computer that has the ability to adapt • A computer that has the ability to reason/make decisions	**3**

10(b)	Any **three** from For example: • It could use machine learning • It can collect data about the **players** actions • **Analyse/identify patterns** in the player's movements • … so it can learn how the player moves • … to help predict the next movement of the player • It can adapt its own movements to match the players // It can adapt its own processes • It can store successful/unsuccessful moves • It will learn the most efficient/optimal movements against a player	**3**

Question 11 · May/June 2026 Paper 11, original Q7 (a),(b),(c),(d) [11 marks]

Question	Answer	Marks	Guidance
11(a)	Acoustic sensor	**1**	Mark the first answer given. Sound/noise sensor is TV. Microphone is NE.
11(b)	Any **two** from: • Proximity sensor • Infrared sensor • Light sensor • Magnetic field sensor	**2**	Mark the first answer given.
11(c)(i)	Any **one** from: • It has a mechanical structure/framework • It is programmable • It has the ability to move	**1**	Mark whole answer. MP1 – It is mechanical is TV MP2 – It can follow instructions is TV

11(c)(ii)	Any **three** from: • The microprocessor/it compares the received data/data sent to the stored value/preset value (of 10) • If the **data/value** received is **less than 10**, (the microprocessor) sends a **signal** to stop the robot • (The signal is sent) to an actuator that is used to stop the robot	**3**	MP2 – 'when an object/distance/robot is less than 10cm' is TV. MP3 – just reference to an actuator is NE here. Must be saying the actuator is what is used to trigger the action to stop the robot in some way.
11(d)	Any **four** from: For example: • It can use machine learning • … by adapting its own data/rules/processes • It can **create/makes** a (digital) map (of the room) // Records/collects data about the **shape/dimensions** of the room • … and **record/store** the **location** of objects • **GPS** may be used to record the location of objects • It can **update** its (digital) map • … to **record/store** when objects are **moved/removed/added** • It can store successful/unsuccessful routes/actions (that avoid objects)	**4**	Do not accept 'it can learn where objects are placed in the room'. This is given in the question.

Question 12 · May/June 2026 Paper 13, original Q6 (a),(b),(c),(d) [7 marks]

Question	Answer	Marks	Guidance
12	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.		

Annotation	Meaning
✓	Correct point
✗	Incorrect point
?	Unclear response
FT	Follow through
REP	Repetition

I	Ignore
BOD	Benefit of doubt given
NBOD	Benefit of doubt not given
TV	Content of response too vague
NAQ	Not answered question
Annotation	**Meaning**
λ	Omission
	Section not relevant
	Section incorrect
Highlighter	Information copied from the text
SEEN	Page or response seen by examiner

12(a)	Any **one** from: • It collects data • It applies rules to the data • It has the ability to reason	**1**	
12(b)	A program that has the ability to **automatically** adapt its **own** data and processes	**1**	
12(c)	**Two** from: For example: • It could **analyse** the **input** actions from the player • It could **analyse** past extra **characters** that performed well • Create/adapt the actions of an extra character based on past characters/gameplay	**2**	

12(d)	Any **three** from: • It takes/uses the data input by the user about the problem • … search for similar knowledge in the knowledge base • … applies it to the rule base (to find the solution) • … decide which question to ask next (if needed)	**3**	